Physics **Basic F = ma Problems** _____

Name Per Date Row

F = m**a** **F** is net force **a** is acceleration in the same direction as the net force

m is mass in Kilograms. Use Newtons as the force unit.

Please Show All Work. Use correct units and significant figures.

1. A rabbit with a mass of 2.0 Kg accelerates at 2.00 m/s^2 . Find the net force on the rabbit

2. A rocket with a mass of 4.0 Kg generates thrust of 200 N. Neglecting air resistance what acceleration should we expect?

3. Benny pushes Betty on her skateboard with a force of 150 N. If she accelerates at 3.0 m/s^2 what is her mass?

4. A 2.0 Kg book sits on a table.
   (a) The net vertical force on the book is _____
   (b) The net horizontal force on the book is _____

5. A 1000 Kg car experiences a 2500 N net force while accelerating. The acceleration is: (show work!)

6. While braking a truck experiences a net force of 5000 N. The mass of the truck is 10,000 Kg. What will be the rate of deceleration of the truck?

   (b) If the truck was initially traveling at 20 m/s, how long will it take for the truck to stop?